Semantic layer surge brief

Repository: pretiumdata-dbt-semantic-layer
Purpose: One printable document for a data-engineering surge — onboarding, architecture, progress, roadmaps, and reference/AI.
Does not replace canonical rules; it **orients** readers to them.

PDF: Export this file from your editor or use Pandoc (example: `pandoc SURGE_BRIEF.md -o SURGE_BRIEF.pdf`). Mermaid diagrams may require a Mermaid-capable pipeline.

Table of contents

1. Canonical authority (pointers)
2. Engineer onboarding
3. Architecture overview
4. Progress snapshot — April 2026
5. Vendor onboarding + analytic studies roadmap
6. Jon — PROD handoff roadmap
7. Reference layer and AI

Name note: Governing docs use **Jon** for Snowflake **`TRANSFORM.[VENDOR]`** PROD ownership.

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1. Canonical authority (pointers)

Topic	Path (from repo root)
Operating model (repos, ownership)	`docs/OPERATING_MODEL.md`
Architecture (layers, Jon vs Alex, metric gates)	`docs/rules/ARCHITECTURE_RULES.md`
Schema matrix (object placement, naming)	`docs/rules/SCHEMA_RULES.md`
Migration procedure + legacy DB ban	`docs/migration/MIGRATION_RULES.md`
Task register (`T-*`)	`docs/migration/MIGRATION_TASKS.md`
Docs index	`docs/README.md`
Catalog seed wave order	`docs/CATALOG_SEED_ORDER.md`

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2. Engineer onboarding

Goal: Land in this repo in under a day with the correct **contracts** and **ownership** in mind.

2.1 What this repo is

pretiumdata-dbt-semantic-layer is the **canonical dbt** home for:

- **`TRANSFORM.DEV`** — typed **`FACT_*`**, **`CONCEPT_*`**, vendor **`REF_*`** (until promoted to Jon's **`TRANSFORM.[VENDOR]`**).
- **`REFERENCE.CATALOG`** / **`REFERENCE.DRAFT`** / **`REFERENCE.GEOGRAPHY`** — governed vocabulary and geography spine (see §7).
- **`ANALYTICS.DBT_*`** — **`FEATURE_*`**, **`MODEL_*`**, **`ESTIMATE_*`** (and **`QA_*`** on STAGE), **not** facts or concepts.
- **`SERVING.DEMO`** — dev delivery views (thin contract; see `docs/reference/SERVING_DEMO_ICEBERG_TARGETS.md`).

pretium-ai-dbt remains a **migration source** (SnowSQL, some ingest, legacy patterns) until retired. Closure for migration work is **here**, not only in the sibling repo — see `docs/migration/CANONICAL_COMPLETION_DEFINITION.md`.

2.2 Non-negotiables

1. **Semantics over shortcuts:** governed CSVs (`seeds/reference/catalog/*.csv`) must be **literally correct** (codes, grains, joins), not "close enough." Repo rule: semantics-over-generic-process.
2. **No legacy PROD DBs in this dbt graph:** no reads of **`TRANSFORM_PROD`**, **`ANALYTICS_PROD`**, **`EDW_PROD`** from `models/` / `macros/` / `tests/` (CI-enforced; see `docs/migration/MIGRATION_RULES.md`).

3. **Prefix placement:** `FACT_*` / `CONCEPT_*` only in `TRANSFORM.DEV`. `FEATURE_*` / `MODEL_*` / `ESTIMATE_*` only in `ANALYTICS.DBT_*`. See `docs/rules/ARCHITECTURE_RULES.md`.
4. **Catalog tokens before object names:** every `[concept]`, `[geo_level]`, `[frequency]`, … token in a new object name needs an **active** `REFERENCE.CATALOG` row first (`docs/rules/SCHEMA_RULES.md`).

2.3 Day-zero setup

From repo root (see also root `README.md`):

```
python -m venv .venv && source .venv/bin/activate
pip install dbt-snowflake sqlfluff
dbt deps
```

Profiles: use your org's Snowflake profile; for CI-style parse-only work, read the comments in `ci/profiles.yml` (parse vs `ci` target).

```
dbt debug --target dev   # or your Snowflake target
dbt parse
```

Catalog work: follow `docs/CATALOG_SEED_ORDER.md` strictly.

2.4 Where to work by task type

You are…	Start in…	Also read…
Porting / building vendor facts	`models/transform/dev/{vendor}/`	`docs/migration/MIGRATION_TASKS_*.md`
Unifying market signals	`models/transform/dev/concept/`	`docs/migration/MIGRATION_FACT_SYSTEMIZATION_PLAYBOOK.md`
Analytics / scores / forecasts	`models/analytics/feature`, `model`, `estimate`	`docs/migration/MODEL_FEATURE_ESTIMATION_PLAYBOOK.md`
Registering metrics	`seeds/reference/catalog/metric.csv`	`docs/migration/METRIC_INTAKE_CHECKLIST.md`
QA / gates	`models/analytics/qa/`, `tests/`	`docs/migration/QA_GOVERNANCE_TEST_TYPES.md`

2.5 Logging and "done"

- Append batch history and model moves to `docs/migration/MIGRATION_LOG.md` / `docs/migration/MIGRATION_BATCH_INDEX.md` per field guide in the log header.
- Update `T-*` rows in `docs/migration/MIGRATION_TASKS.md` when disposition changes.

2.6 Who to ask

- **Alex-owned** graph targets and **REFERENCE** seeds: Alex (`docs/rules/ARCHITECTURE_RULES.md`).
- **Vendor PROD cleanse**, `TRANSFORM.[VENDOR]`, `SOURCE_PROD` → `TRANSFORM.[VENDOR]` pipeline: **Jon** — see §6.

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3. Architecture overview

Compressed view of `docs/rules/ARCHITECTURE_RULES.md` and `docs/rules/SCHEMA_RULES.md`.

3.1 Layer cake (data flow)

```
flowchart TB
  subgraph ingest [Ingest — Jon primary]
    SP[SOURCE_PROD.VENDOR RAW_*]
    TV[TRANSFORM.VENDOR cleansed PROD]
  end
  subgraph alex_transform [Alex — TRANSFORM.DEV]
    FACT[FACT_* vendor typed facts]
    REF[REF_* vendor crosswalks dev]
    CONCEPT[CONCEPT_* unions / canonical grains]
  end
  subgraph reference [Alex — REFERENCE]
    GEO[REFERENCE.GEOGRAPHY census spine]
    CAT[REFERENCE.CATALOG seeds]
    DRAFT[REFERENCE.DRAFT staging seeds]
  end
  subgraph analytics [Alex — ANALYTICS.DBT_*]
    FEAT[FEATURE_* views]
    MOD[MODEL_* views]
    EST[ESTIMATE_* tables]
    QA[QA_* STAGE gates]
  end
  SP --> FACT
  TV --> FACT
  FACT --> CONCEPT
  GEO --> FACT
  GEO --> CONCEPT
  CAT --> FACT
  CAT --> CONCEPT
  CAT --> FEAT
  CONCEPT --> FEAT
  FEAT --> MOD
  MOD --> EST
  FEAT --> QA
```

Reading order for new market metrics: vendor landing → (optional Jon cleanse) → `FACT_*` → `CONCEPT_*` → `FEATURE_*` → `MODEL_*` → `ESTIMATE_*` (when registered).

3.2 Ownership split

Snowflake area	Primary owner	Surge engineers
`SOURCE_PROD.[VENDOR].RAW_*`	Ingest / Jon path	No transform logic in landings
`TRANSFORM.[VENDOR]` (PROD vendor silver)	**Jon**	**Read** only; no writes to Jon schemas
`TRANSFORM.DEV` `FACT_*` / `CONCEPT_*` / `REF_*`	**Alex**	Primary migration surface
`REFERENCE.GEOGRAPHY`	**Alex**	Census spine; no vendor xwalks
`REFERENCE.CATALOG` / `REFERENCE.DRAFT`	**Alex**	Vocabulary + metric registry
`ANALYTICS.DBT_*`	**Alex**	Features, models, estimates, QA
`TRANSFORM.FACT` (future)	**Jon**	Alex does **not** create today

3.3 Why `REFERENCE.CATALOG` sits in the middle

Object names embed **tokens** (`concept_code`, `geo_level_code`, `frequency_code`, …). The catalog is the **compiler symbol table** for the warehouse.

Metric registration gates: see `docs/rules/ARCHITECTURE_RULES.md` (Metric Registration Gates).

3.4 Geography discipline

- **Vendor-specific** crosswalks → `TRANSFORM.DEV` `REF_*` until Jon promotes to `TRANSFORM.[VENDOR]`.
- **Non-vendor** census spine → `REFERENCE.GEOGRAPHY` only.

3.5 AI-facing surfaces (conceptual)

- **Today:** Ground definitions in `REFERENCE.CATALOG` (`metric.definition`, concepts). §7.
- **Target:** `REFERENCE.AI` for prompt fragments — design: `docs/reference/OFFERING_INTELLIGENCE_CANON.md`.

3.6 Deep dives by role

Role	Next reads
Fact builder	`docs/migration/MIGRATION_FACT_SYSTEMIZATION_PLAYBOOK.md`, vendor `MIGRATION_TASKS_*.md`
Analytics	`docs/migration/MODEL_FEATURE_ESTIMATION_PLAYBOOK.md`, `docs/migration/PLAYBOOK_ANALYTICS_FEATURES_FROM_CATALOG.md`
Catalog / metrics	`docs/migration/MIGRATION_TASKS_VENDOR_METRIC_CATALOG_INTAKE.md`, `docs/CATALOG_SEED_ORDER.md`

4. Progress snapshot — April 2026

Scope: Repo-local file counts. `T-*` status: `docs/migration/MIGRATION_TASKS.md` and `docs/migration/MIGRATION_LOG.md`.

4.1 Reference catalog (governed seeds)

Artifact	Location	Approx. rows (excl. header)
Vendors	`seeds/reference/catalog/vendor.csv`	53
Datasets	`seeds/reference/catalog/dataset.csv`	89
Concepts	`seeds/reference/catalog/concept.csv`	31
Metrics	`seeds/reference/catalog/metric.csv`	~4,969
Product type ↔ metric bridge	`seeds/reference/catalog/bridge_product_type_metric.csv`	~648
Derived analytics metrics	`seeds/reference/catalog/metric_derived.csv`	7 (active / planned)

More seeds: `docs/CATALOG_SEED_ORDER.md`.

4.2 Transform DEV — facts and concepts

Pattern	Approx. count	Notes
`FACT_*.sql` under `models/transform/dev/`	**106**	Government, Zillow research, fund/OpCo, labor risk, corridor, etc.
`CONCEPT_*.sql` under `models/transform/dev/`	**27**	Market-monthly unions, OpCo progress

Migration log: `MIGRATION_LOG.md` summary shows **57 migrated** / **81** tracked legacy FACT paths, **24 pending** (named lineage in log).

4.3 Analytics layer

Pattern	Approx. count	Notes
`FEATURE_*.sql`	**10**	
`MODEL_*.sql`	**4**	
`ESTIMATE_*.sql`	**0** in `models/analytics/estimate/`	Contract defined; models not shipped yet
`QA_*.sql`	**25**	

4.4 Portfolio breadth (legacy dataset lens)

`docs/migration/MIGRATION_REGISTRY_VENDORS_DATASETS_METRICS.md` §3: **pretium-ai-dbt** `dim_dataset_config.csv` — **147 datasets** / **63 vendor IDs**. This repo's `dataset.csv` is the **governed** subset.

4.5 Next milestones

1. Close **`T-VENDOR-*-READY`** tasks with smoke SQL + catalog rows.
2. **`T-CATALOG-METRIC-VENDOR-ROLLOUT`** (`in_progress`) — grow `metric.csv` with architecture gates.
3. First **`ESTIMATE_*`** under `models/analytics/estimate/` when `metric_derived` promotes — `docs/migration/MODEL_FEATURE_ESTIMATION_PLAYBOOK.md`.

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5. Vendor onboarding + analytic studies roadmap

5.1 Vendor onboarding

Status: `docs/migration/MIGRATION_TASKS.md`
Registry: `docs/migration/MIGRATION_REGISTRY_VENDORS_DATASETS_METRICS.md` §2.

Theme	Representative tasks	Checklist
ApartmentIQ + Yardi Matrix	`T-VENDOR-APARTMENTIQ-READY`, `T-VENDOR-YARDI-MATRIX-READY`	`docs/migration/MIGRATION_TASKS_APARTMENTIQ_YARDI_MATRIX.md`
Yardi operational (BH / Progress)	`T-VENDOR-YARDI-READY`	`docs/migration/MIGRATION_TASKS_YARDI_BH_PROGRESS.md`
First Street + RCA / MSCI	`T-VENDOR-FIRST-STREET-READY`, `T-VENDOR-RCA-READY`	`docs/migration/MIGRATION_TASKS_FIRST_STREET_RCA.md`
CoStar	`T-VENDOR-COSTAR-READY`	`docs/migration/MIGRATION_TASKS_COSTAR.md`
Cherre	`T-VENDOR-CHERRE-READY`	`docs/migration/MIGRATION_TASKS_CHERRE.md`
Redfin + Stanford SEDA	`T-VENDOR-REDFIN-READY`, `T-VENDOR-STANFORD-READY`	`docs/migration/MIGRATION_TASKS_STANFORD_REDFIN.md`
BPS / Census / BLS / LODES	`T-TRANSFORM-*`	`docs/migration/MIGRATION_TASKS_TRANSFORM_BPS_CENSUS_BLS_LODES.md`
Corridor pipeline	`T-CORRIDOR-*`	`docs/migration/MIGRATION_TASKS_CORRIDOR_PIPELINE_SOURCES.md`
Cybersyn	`T-CYBERSYN-*`	`docs/migration/MIGRATION_TASKS_CYBERSYN_SOURCE_SNOW.md`, `docs/reference/CYBERSYN_GLOBAL_GOVERNMENT_BRING_IN_MATRIX.md`
Zillow	`T-*` in doc	`docs/migration/MIGRATION_TASKS_ZILLOW_TRANSFORM_DEV.md`

Bulk path: ~**2,129** pretium-ai-dbt model paths (`MIGRATION_TASKS_INVENTORY_models.txt`); fact gap waves: `docs/migration/MIGRATION_FACT_GAP_WAVE_QUEUE.md`.

Vendor context: `docs/vendor/vendors.md`, `docs/vendor/0_inventory/`.

5.2 Planned studies (validation / governance)

Maps to **GOV_*** taxonomy: `docs/migration/QA_GOVERNANCE_TEST_TYPES.md`.

Catalog / metrics: `docs/migration/QA_METRIC_LAYER_VALIDATION.md`, `docs/migration/QA_TRANSFORM_DEV_CATALOG_REGISTRATIONS.md`, `docs/migration/METRIC_INTAKE_CHECKLIST.md`.

QA model themes (`models/analytics/qa/`): concept–feature parity (`qa_feature_concept_parity_diff.sql`, `qa_feature_concept_alignment.sql`); collinearity / series collision; geography sensitivity / cold start; distribution shift / structural breaks; vendor blend ablation (`qa_vendor_blend_ablation_rent.sql`).

Cross-cutting: `docs/migration/VENDOR_CONCEPT_COVERAGE_MATRIX.md`, `docs/migration/MIGRATION_TASKS_SOURCES_GAP_ANALYSIS.md`, `docs/migration/MIGRATION_TASKS_POLARIS_DATASET_PRIORITIES.md`, `docs/migration/MIGRATION_TASKS_STRATA_BACKEND_LINEAGE.md`, `docs/migration/MIGRATION_TASKS_TEARSHEET_SERVICE.md`.

Example end-to-end slice: `docs/migration/LABOR_AUTOMATION_RISK_STACK_SEMANTIC_LAYER.md`, `docs/runbooks/RUN_LABOR_AUTOMATION_RISK_STACK_DBT.md`.

5.3 How to use during the surge

1. Pick vendor cluster → `MIGRATION_TASKS_*.md` → inventory SQL → `MIGRATION_LOG.md` on exit.
2. Pair each FACT wave with a **GOV_*** study.
3. Do not mark `T-* migrated` without catalog alignment when metrics matter — `T-CATALOG-METRIC-VENDOR-ROLLOUT`.

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6. Jon — PROD handoff roadmap

Audience: Jon — `TRANSFORM.[VENDOR]` and `SOURCE_PROD` **→ vendor silver** (`docs/rules/ARCHITECTURE_RULES.md`, `docs/OPERATING_MODEL.md`).

Alex: `TRANSFORM.DEV`, `REFERENCE.*`, `ANALYTICS.DBT_*` — **read** Jon PROD; **no** dbt writes to `TRANSFORM.[VENDOR]`.

Many `FACT_*` models read `SOURCE_PROD RAW_*` until `TRANSFORM.[VENDOR]` exists; **end state** is `ref()` **/** `source()` to Jon-cleansed tables. Vendor **crosswalks** move from Alex `REF_*` to `TRANSFORM.[VENDOR]` when promoted.

6.1 Promotion sequence

Step	Jon action	Unblocks
J0	`SOURCE_PROD RAW_*` contracts stable	Alex `source()` compiles
J1	Cleanse silver in `TRANSFORM.[VENDOR]`	Alex re-points `FACT_*` off RAW-only
J2	Vendor xwalks in `TRANSFORM.[VENDOR]`	Metric geo compliance
J3	Grants, PKs, cadence, breaking-change policy	QA, serving
J4	Optional future `TRANSFORM.FACT`	Long-term canonical facts (Jon-owned; not Alex-created today)

6.2 Vendor clusters (typical PROD dependency)

Live list: `MIGRATION_TASKS.md`. Examples: ApartmentIQ, Yardi Matrix, Yardi BH/Progress, CoStar, Cherre, First Street, RCA/MSCI, Redfin, Stanford; government feeds often `TRANSFORM.BLS` **/** `CENSUS` **/** `LODES` **/** `BPS`.

6.3 Geography boundary

Vendor geo normalization → `TRANSFORM.[VENDOR]` or temp `TRANSFORM.DEV REF_*`. Do not duplicate `REFERENCE.GEOGRAPHY` in vendor schemas without coordination.

6.4 Communication protocol

1. Breaking table renames: old/new FQN before cutover.
2. After Jon promotes silver: Alex PR to switch reads + `MIGRATION_LOG.md`.
3. `metric.table_path` stable before metric registration finalizes.

6.5 Jon-facing artifacts

`docs/migration/artifacts/batch012_yardi/`, `docs/migration/MIGRATION_BASELINE_RAW_TRANSFORM.md`, `docs/rules/TRANSFORM_VENDOR_DESIGN_PRINCIPLES.md`.

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7. Reference layer and AI

7.1 Two meanings of "reference"

Meaning	Where	What
A) Repo `seeds/reference/`	Git CSV + YAML	**Authoring** → becomes Snowflake `REFERENCE.*` via `dbt seed` / build.
B) Snowflake `REFERENCE.CATALOG`, `REFERENCE.GEOGRAPHY`, `REFERENCE.DRAFT`, (future) `REFERENCE.AI`	Snowflake	**Runtime** registry for dbt, QA, apps, prompts.

7.2 `seeds/reference/catalog/` → `REFERENCE.CATALOG`

Controlled vocabulary and semantic registry: `concept.csv`, `vendor.csv`, `dataset.csv`, `metric.csv` (canonical metric SoT), dimension seeds (`geo_level`, `frequency`, `model_type`, `estimate_type`, …), bridges (`bridge_product_type_metric.csv`), `metric_derived`, `metric_derived_input`, `catalog_wishlist.csv`. Load order: `docs/CATALOG_SEED_ORDER.md`.

7.3 `seeds/reference/draft/` → `REFERENCE.DRAFT`

In-progress rows not yet promoted to catalog.

7.4 Geography → `REFERENCE.GEOGRAPHY`

`models/reference/geography/` — census spine, H3, ZIP–county–CBSA; **no vendor-specific** xwalks.

7.5 Uses in the stack

Governance tests; naming tokens before new objects; authoritative definitions for humans and tools; `QA_*` vs catalog expectations.

7.6 AI (practical)

- **Today:** RAG / agents use `REFERENCE.CATALOG.metric` (`definition`, grains, `table_path`) plus concept / dataset / vendor rows as **ground truth**. Align `FEATURE_*` / `MODEL_*` with `metric_derived` when active (`docs/reference/CATALOG_METRIC_DERIVED_LAYOUT.md`).
- **Target:** `REFERENCE.AI` for governed prompt/lens text — `docs/reference/OFFERING_INTELLIGENCE_CANON.md`; `seeds/reference/ai/` not fully present yet.
- **Avoid** legacy FQNs in new prompts (`EDW_PROD`, retired catalog paths).

One line: `REFERENCE.CATALOG` is the dictionary and symbol table; `REFERENCE.GEOGRAPHY` is the shared map; **AI** should read them before inventing metric meaning; `REFERENCE.AI` will hold governed narrative lenses on top.

End of surge brief. Split chapters for editing: `docs/surge/ENGINEER_ONBOARDING.md`, `ARCHITECTURE_OVERVIEW.md`, `PROGRESS_SNAPSHOT_2026-04.md`, `VENDOR_AND_ANALYTIC_STUDIES_ROADMAP.md`, `JON_PROD_HANDOFF_ROADMAP.md`, `REFERENCE_AND_AI.md`.